Part II, Lecture 2: Agencies' and Internal Ratings, Judgement-Based and Statistical-Based Ratings

By the end of this session you must know why ratings matter across capital markets, lending and regulation, how agency, internal and market-implied ratings differ in producer and time horizon (TTC vs PIT), the dimensions and processes matrix (issuer vs issue, assignment vs quantification), how judgment-based assignment works at agencies and why ratings can fail, how statistical-based internal ratings are built (modular architecture, reduced-form discriminating models, cut-offs and rating bands), and the time-frame choices regulators expect for assignment and quantification.

Plan

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1. Why Ratings Matter

Credit ratings inform investors and facilitate bond issuance and syndicated loans in capital markets, they feed credit risk quantification in risk management, they guide credit decisions, pricing and portfolio management, they drive the cost of capital for borrowing entities, and they sit inside banking regulation, for example the Basel framework.

2. Three Producers, Three Philosophies: TTC vs PIT

Agencies' ratings: produced by S&P, Moody's, Fitch. Typically Through-the-Cycle (TTC): they aim to measure risk across the economic cycle, stable over time.

Internal ratings: produced by banks for their own lending. Tend to have a high degree of PIT-ness (Point-in-Time): they reflect current conditions and behavioural data.

Market-implied ratings: derived from market data (credit spreads, CDS spreads, equity prices) transformed into a default-risk indicator. Very PIT oriented, can be volatile.

3. Agencies' Ratings

- Long history: from 19th century mercantile credit agencies, through John Moody rating US railroad bonds in 1909, to the 1941 merger forming Standard & Poor's. Today's major players: S&P, Moody's, Fitch.
- Significant for capital markets (bonds, syndicated loans), with historical evidence of predictive power and methodologies codified over time.
- Access to issuers' internal information; solicited vs unsolicited ratings exist; reputation is the key asset justifying why investors demand ratings and issuers pay for them.
- Role in Basel II: agencies are recognised as ECAIs for banks using the Standardized Approach.

4. Internal Ratings

- Developed from the mid-1990s, strongly boosted by Basel II's Internal Ratings-Based (IRB) approaches.
- Used by banks for their own lending decisions, pricing and capital calculation; confidential to the bank.
- Higher PIT component than agencies' ratings (current conditions, behavioural data).
- Natural for SME financing: proximity advantage, access to detailed information, cost recovery through loan margins.

5. Market-Implied Ratings

- Spreading since the 1990s; built by transforming market data into a default-risk indicator.
- The conversion relies heavily on assumptions that supplement the calculations; the models are relatively simple.
- Very Point-in-Time: reflect current market sentiment and liquidity conditions; low stability over time.

6. The Dimensions and Processes Matrix

Must know: ratings vary along two dimensions (issuer/borrower vs issue/facility) and two processes:
• **Assignment** (ordinal output): issuer/borrower ratings or issue/facility ratings
• **Quantification / calibration** (cardinal output): for borrowers, ex ante probabilities of default and ex post default rates; for facilities, ex ante expected loss rates and ex post loss rates

Comparing ratings across different systems requires a master scale mapping each system's classes onto one common scale.

7. Judgment-Based Assignment and Rating Failures

Agencies primarily use judgment-based (principles-based) assignment:

- Not mechanical: a rating is, in the end, an opinion. Complex process: preliminary analysis, meetings with the issuer, analyst presentation, rating committee discussion and vote.
- Wide information range: objective and subjective, quantitative and qualitative, historical and forward-looking, including sector analysis, competitive analysis, management quality. Financial ratios inform but do not rigidly determine the rating.
- Surveillance: ongoing monitoring, annual reviews, triggers for action (CreditWatch, outlook change, rating change).

Possible causes of ratings failure (Bhatia 2002), to know cold:

Must know:
• **Information risk:** inaccurate data provided by the rated institutions
• **Analytical constraints:** agencies may lean on already issued rating reports to overcome scarce human resources
• **Revenue bias:** issuers may pay additional fees hoping for a more favorable rating
• **Other incentive problems:** conflicts of interest from other services offered, for example risk management advice
• Balanced against these: **reputation**, the only asset that justifies investors demanding ratings and issuers paying for them

8. Statistical-Based Assignment: Models, Cut-Offs, Bands

Banks primarily use statistical-based assignment for internal ratings:

- Models interpret historical quantitative data: financial statements, internal behavioural data, credit bureau, credit register. They produce a model-based rating. For retail or simple segments this is often the final rating, no override.
- Increasingly enriched by new data (internet) and methods (ML/AI), with regulatory caution about black boxes.

Typical architecture: separate modules per data source (financial scoring, behavioural scoring, credit register or bureau scoring, business sector rating, questionnaire-based qualitative rating) feed a model-based rating, then an override step produces the borrower rating, then the facility rating with its LGD rating. Modules facilitate maintenance, allow recalculating one source, give a clearer customer profile, handle missing data, and allow specialized techniques.

Estimation process (reduced-form models):

1. Dataset of borrowers at a "time zero", including future defaulters and performers.
2. Potential explanatory variables: only data available before the outcome is known.
3. Dependent variable: the outcome (0 performing, 1 defaulted).
4. Techniques: stepwise selection for variables, Linear Discriminant Analysis or logistic regression for weights.
5. Result: a score Z = a1 x X1 + a2 x X2 + … that best discriminates between defaults and non-defaults, minimizing both Type I and Type II errors.

Two-variable example: Z = a x ROE + b x ROA. Choosing a, b and a cut-off Z* defines a separating line ROE = (Z*/a) - (b/a) x ROA. *To assign ordinal rating classes rather than a binary accept/reject, set multiple cut-offs Z1*, Z2, *Z3*, …: parallel lines (same slope, different intercepts) creating rating bands. Higher score means higher predicted quality and lower expected default rate.

The balance between automated and labor-intensive techniques varies with the borrower's complexity and the loan's gross contribution: statistical algorithms (for example discriminant scoring) for installment loans and small business, human-based procedures for qualitative aspects of large corporate borrowers.

9. Time-Frame Decisions

- Common observation period for PD estimation: one year. Longer horizons make models more forward-looking but reduce predictive power; behavioural data lose relevance as the horizon lengthens.
- Regulatory expectation (Basel II paragraph 414, ECB Guide 2024): rating assignment should target a longer time horizon (2-3 years per the ECB), while PD quantification stays at one year.
- The link between time-zero data and observation-period outcome must be consistent for all borrowers in the dataset.

- Time-stratification: using different time zeros for different borrowers reduces dependence on a single calendar year or cycle stage, smoothing the model’s relation to the cycle.

10. Common Mistakes and Exam Traps

- Assigning TTC philosophy to every rating type. Agencies are typically TTC, internal ratings lean PIT, market-implied ratings are very PIT.
- Confusing the matrix cells: borrower quantification targets probabilities of default, facility quantification targets expected loss rates.
- Treating financial ratios as mechanically determining an agency rating. They inform a committee’s opinion, they do not decide it.
- Forgetting that revenue bias and conflicts of interest are only two of the four failure causes; information risk and analytical constraints are independent problems.
- Believing the model-based rating is always final. In retail segments it often is, but the architecture includes an override step for complex borrowers.
- Mixing the horizons: assignment targets a longer horizon (2-3 years), quantification stays at one year. Reversing them contradicts Basel II and the ECB Guide.

11. Must Know Absolutely

Must know:
• Three producers: agencies (TTC), banks’ internal ratings (PIT-leaning, IRB-boosted, confidential), market-implied (very PIT, assumption-heavy, volatile)
• Matrix: issuer vs issue dimensions, assignment (ordinal) vs quantification (cardinal: PD for borrowers, expected loss rates for facilities)
• Four causes of rating failure per Bhatia 2002: information risk, analytical constraints, revenue bias, incentive conflicts; reputation is the counterweight
• Statistical models: modules per data source, reduced-form score Z = sum of ai x Xi, cut-offs create parallel rating bands, goal is minimizing Type I and II errors
• Time frames: assignment targets 2-3 years (Basel II 414, ECB 2024), PD quantification 1 year; time-stratification smooths cycle dependence